

ALI NIKKHAH

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SUMMARY

AI Engineer with 2+ years designing and deploying production agentic systems, LLM-powered backends, and intelligent retrieval infrastructure. At Digikala, led the enterprise AI intelligence layer serving 5M+ daily queries using stateful LangGraph multi-agent pipelines, hybrid RAG (FAISS + Elasticsearch via RRF), HyDE query expansion, KAG symbolic reasoning, and async FastAPI with vLLM and Triton. At Turquoise Digital, engineered a financial agentic chatbot with real-time Whisper/wav2vec2.0 voice transcription, secure function-calling for live portfolio queries, and a time-series transformer portfolio allocation system.

TECHNICAL SKILLS

Languages: Python (Asyncio, Pydantic), Go, C++, SQL, Bash, TypeScript
Deep Learning & AI: PyTorch, TensorFlow, JAX, Hugging Face, LangGraph, LangChain, LlamaIndex, FAISS, HyDE, KAG
Vision & Audio: OpenCV, ViT, BLIP, T5, Whisper, wav2vec2.0, TensorRT, TensorFlow Lite
Data & Streaming: Spark, Trino, Airflow, Kafka, PeerDB, Elasticsearch, Parquet
MLOps & Infra: Docker, Kubernetes, Terraform, Ray, Triton, vLLM, MLflow, Prometheus, Grafana

PROFESSIONAL EXPERIENCE

Digikala

Tehran, Iran

AI Engineering R&D Lead

Jan 2025 – Aug 2025

- Led architectural transition from static semantic matching to stateful **LangGraph** multi-agent pipelines now handling **5M+ daily queries** (p99 latency <200ms) across product discovery, customer support, and internal knowledge retrieval; designed a Modular Control Pipeline (MCP) decoupling ingestion, chunking, retrieval, and synthesis phases for zero-downtime LLM hot-swapping.
- Built hybrid retrieval combining dense **FAISS** vector search with sparse **Elasticsearch** keyword lookup fused via Reciprocal Rank Fusion (RRF); added **HyDE** query expansion and hierarchical chunking to close accuracy gaps on ambiguous long-tail queries (**+23% recall@10**), and deployed **KAG** symbolic reasoning to reduce hallucinations on structured catalog data (**-41% factual error rate**).
- Engineered async **FastAPI** serving with multi-tier **Redis** caching and request batching; deployed LLMs via **vLLM** (PagedAttention) and **Triton Inference Server** to maximize KV-cache utilization and GPU throughput (**3.2x throughput gain, 60% cost reduction per 1K queries**) under burst traffic.

Digikala

Tehran, Iran

AI Engineer

Nov 2024 – Jan 2025

- Prototyped RAG architectures against production query logs; integrated LangGraph cyclic agent graphs for multi-turn session management, context re-evaluation, and automated query rewriting; benchmarked retrieval precision and latency to shape the R&D roadmap for the lead role.

Turquoise Digital (Firouzeh Digital)

Tehran, Iran

Senior Data Engineer & Data Scientist

Sep 2025 – Mar 2026

- Engineered an enterprise **LangGraph** financial chatbot with real-time **Whisper/wav2vec2.0** call transcription and secure dynamic function-calling for live equity balance queries and compound-profit calculations — without exposing database credentials to the LLM context (**-78% ticket volume, 94% transcription accuracy**).
- Built an AI portfolio allocation system fusing historical time-series pricing with streaming news-sentiment embeddings using time-series transformer models, producing forecasts with explicit confidence intervals that drive automated batch trading allocations (**Sharpe +0.42 vs benchmark**).

HomaCloud

Tehran, Iran

MLOps Engineer

Aug 2021 – Feb 2022

- Orchestrated distributed ML training and feature-extraction pipelines on Kubernetes (Ray, MLflow); built FastAPI inference APIs with TensorFlow Serving; automated the full CI/CD lifecycle with Docker, ArgoCD, and GitHub Actions, cutting deployment cycles to under 10 minutes (**15+ models, 99.9% deploy success rate**).

EDUCATION

Sharif University of Technology

Tehran, Iran

B.Sc. Electrical Engineering – Digital Systems GPA: 3.65 overall, 3.91 major

Sept 2020 – Jul 2024

RESEARCH HIGHLIGHTS

Univ. of British Columbia (*Remote RA, Feb 2024 – Nov 2025*) — Vision-language architectures (ViT + BLIP + T5) with self-supervised contrastive learning for automated DICOM-compliant ultrasound report generation; achieved SOTA BLEU & ROUGE-L. Preprint in preparation.

L3S Research Center (*Remote RA, Apr 2023 – Feb 2024*) — 3D CNNs and transformer encoders over dense optical flow for texture-free motion classification; SOTA on UCF-101 and HMDB-51.